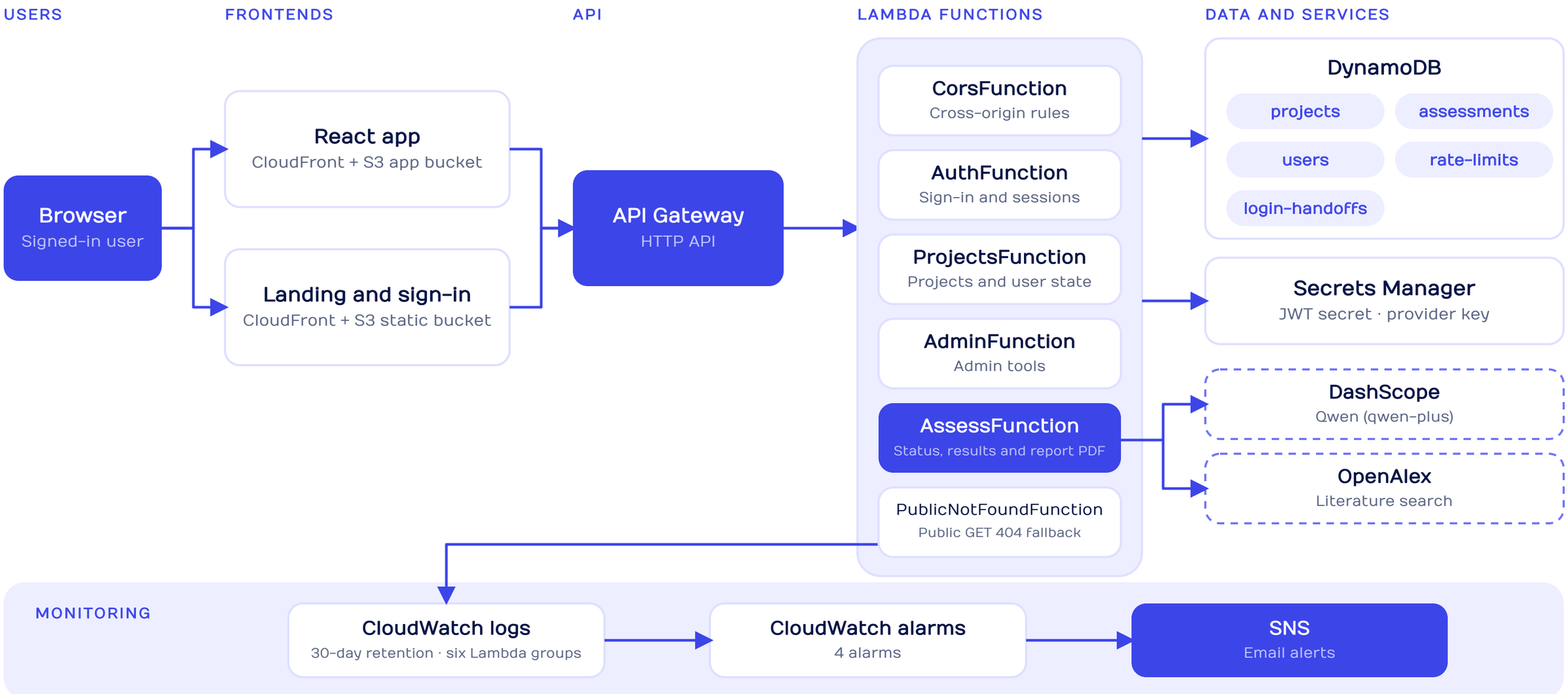
plusquam

ARCHITECTURE

System architecture

How the Plusquam staging environment is put together.



Dashed outline = external service

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DATA FLOW

Data and request flow

From a new project to a finished report.

INPUT

Sign in
Standard or admin account

→

Create project
Start a new assessment

→

Optional PDF or DOCX
Validated, text extracted and saved on the project

↓

ASSESSMENT · ASSESSFUNCTION

1. **Query understanding**
2. **Legal context**
3. **Literature retrieval** — OpenAlex
4. **Research synthesis**
5. **Open-question identification**
6. **Novelty scoring** — Five dimensions
7. **Assessment compile and store** — Saved to DynamoDB

↓

Report saved
Stored in DynamoDB

→

Results page
Owner and admin views

→

PDF on request
Generated on request

Qwen and literature search

AssessFunction calls DashScope qwen-plus (temperature 0.3). Literature comes from OpenAlex.

Uploaded files

The original file is not stored. Only the extracted text is saved on the project record.

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SECURITY

Security posture at a glance

All nine findings fixed and verified on staging.

9/9
findings fixed

11/11
checks passed

0
known dependency vulnerabilities

4
alarms monitoring the system

FINDINGS BY SEVERITY



High 1 · Medium 4 · Low 4

11 CHECKS, ALL PASSED

- Authentication
- CORS allowlist
- Session revocation
- Unapproved origin
- Role enforcement
- Browser secret check
- Login rate limit
- Security scan (ZAP)
- Cross-user access
- Data at rest
- S3 lockdown

DATA PROTECTION AND SCAN

AES256	Both S3 buckets, public access blocked
KMS	All five DynamoDB tables use KMS encryption
PITR	Projects, assessments and users tables only
ZAP	0 High · 0 Medium · 0 Low alerts

FINAL STATUS

All nine findings are fixed and verified on staging. Access control, rate limiting, upload rejection, data encryption, scans and alert delivery all passed.

UAT results dashboard

Every scenario was run on staging with synthetic test data.

12	0	None recorded
scenarios passed	scenarios failed	defects

12 of 12 passed

UAT-01 PASS Sign in and reach the workspace	**UAT-02** PASS Create a new project	**UAT-03** PASS Upload and read a valid document	**UAT-04** PASS Run an assessment and follow progress
UAT-05 PASS Qwen returns a usable assessment	**UAT-06** PASS Review the results page	**UAT-07** PASS Find a project in History	**UAT-08** PASS Download the report PDF
UAT-09 PASS Review admin records and actions	**UAT-10** PASS Check incomplete and unsuitable input	**UAT-11** PASS Prevent accidental duplicate runs	**UAT-12** PASS Qwen integration and quality check